

to tents, parachutes or camouflage nets.

e-garments are already a big part of the US army. These are the main element of their next-generation soldier system.

e-textile fibres woven as mattresses in NASA space shuttles or space stations can make the life of space travellers better. Travellers will be able to breathe with less difficulty. e-fibres on their space suits can even pump out body heat and increase the rate of flow of fresh air.

e-clothes are a boon in healthcare as well. These are highly useful in rehabilitation also. An e-cloth worn by a patient can send his vital information like blood pressure and body temperature to a remote device like PDA, allowing doctors to monitor the patient in real time.

e-garments can serve to enrich the life of disabled people. Take the example of LuMax sonar ultrasonic range finder. It works on ultrasonic waves and LilyPad Vibe board. It alerts the blind to stop when there is a solid object in their way by giving a beep sound. Sonar technology along with adaptive technology can lead to a lot of e-textile miracles. A sonic realm jacket designed by Threads group at University of Michigan is a great help to the hearing impaired. It consists of a series of vibrating motors kept along spine. These motors translate sound into tactile information.

In homes, e-clothes can act as your personal assistant. Their main purpose is to improve your quality of life. Take the example of a wallet with e-fabric over it. This wallet RFID-linked to a purse can make the purse beep if you are leaving home without the wallet. A piece of e-cloth on the door can provide an alerting beep sound if you forget to lock the door while leaving home.

In stage shows, light control, music and apparatus of the artists are key factors. e-textiles easily ensure light control, illusion and communication required for stage shows. The first e-textile-based stage show was held in

TABLE I
Comparison of Different Materials Used in e-textile Fabrics and Components

Characteristics	Metals and semiconductors	Organic electronic materials
Durability	More	Less
Bending	More	Less
Stiffness	More	Less
Washability	More	Less
Ease of manufacturing	Less	More
Cost	More	Less
Camouflage	Less	More

1990 by a group of students from MIT Massachusetts. They developed artistic applications for electronics, such as a micro music playing system and apparels that change colour when their sensors are alerted.

There are many new ideas coming in, especially in the area of mobile phones. A team of scientists from South Carolina, headed by Xiaodong Li, is busy developing a t-shirt that can recharge mobile phones. This t-shirt consists of an interwoven piece of e-cloth with its yarns acting as electrodes. With these electrodes, the researchers were able to turn the fabric into several capacitors capable of storing electric charge. The circuit of the e-cloth even consists of micro sensors that can absorb body heat and heat of the room.

Material aspects

There are innumerable materials, from conductors to insulators, and from magnetic to non-magnetic. Materials that meet most of the demands of e-textile components should be selected.

Piezoelectric materials are of great use in e-textiles. A flexible memory fabric can be made of platinum and woven copper and copper oxide wires that form the memory circuit.

One of the focus areas of e-textile research is better material for e-yarns. Presently, an e-textile group is investigating a special polymer fibre for different uses. These polymers can act as batteries or as chemical sensors. They can even change their colour for

TABLE II
Specifications of Arduino LilyPad

Technical specifications	Details
Microcontroller	ATmega168V or ATmega328V
Operating voltage	2.7-5.5V
Input voltage	2.7-5.5V
Digital input/output (I/O) pins	14
Analogue I/O pins	6
Flash memory	16 kB (of which 2 kB used by bootloader)
SRAM	1 kB
EEPROM	512 bytes
Clock speed	8 MHz
Programming	With Arduino software
Power	Can be powered using USB connection or with an external supply
Shape	Circle, 50 mm in diameter and 0.8mm thick

disguise, helping soldiers to deceive enemy troops or create illusion.

Way to go

There is vast scope for research in various segments of e-textiles, including computing architecture and communication networks in fabrics.

Not just the hardware is enough. You need better software to operate them. So the researchers are working to develop software services necessary for almost all e-textile applications. Modelling and simulation tools are also underway.

Use of embedded system and distributed computing is expected to significantly improve e-textile applications. Embedding electronic components like sensors and microcontrollers in fabrics is sometimes called fibre electronics.

For many technological revolutions to be successful, implementation of present trends is largely required. That is where e-textile is heading for. Optimising energy usage, solar technology, hybrid energy systems, touchbuttons and screens are some of the hottest areas of research in e-textile segment also. ●

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